

Juan David Muñoz Bolaños

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jmunozbolanos@gmail.com

May 7, 2026

AT&S Austria Technologie & Systemtechnik Aktiengesellschaft

Fabriksgasse 13

8700 Leoben, Austria

Application for Junior Engineer R&D Prototyping & Quality | Ref. 1385609133

Dear AT&S Recruitment Team,

As an engineer in optical metrology and R&D prototyping, my career spans the spectrum from research to hands-on instrumentation. Following my Bachelor degree in Physics Engineering, I gained ten months of industrial experience as a Junior Machine Vision Engineer at INTECOL S.A.S., where I deployed automated inspection systems for high-speed production lines. This practical foundation inspired me to deepen my expertise through a Master of Science in Photonics in Jena and a PhD at the Medical University of Innsbruck, where I developed optical instrumentation integrating optics, fine mechanics, electronics, and software. This path aligns with the innovation and high-tech manufacturing culture at AT&S.

A central focus of my PhD has been executing Design of Experiments and performing equipment qualification with precision research environments. I built closed-loop wavefront sensing platforms from scratch, achieving diffraction-limited imaging performance confirmed in peer-reviewed publications. My participation in the Summer Schools of ASML and ZEISS SMT provided insights into the high precision requirements and quality standards of the semiconductor industry, preparing me to support your R&D trials, metrology equipment operations, and prototype manufacturing activities.

Having lived in Innsbruck for several years, Austria has become my home. As I complete my PhD in June 2026, I look forward to contributing to and growing with your team in Leoben for the long term. I am especially motivated by the opportunity to apply my research and prototyping expertise in an international industrial setting. My Austrian residence permit transitions seamlessly into full work authorization starting May 2026.

I would be grateful for the opportunity to discuss my qualifications in an interview.

Sincerely,

Juan David Muñoz Bolaños

Juan David Muñoz Bolaños

Optical System Engineer

Rechenauerstraße 42, Innsbruck, Austria

+43 676 9115145

munozbolanos@gmail.com | LinkedIn Profile



Summary

Optical System Engineer with a background in Physics and in R&D prototyping. Closing PhD program in June 2026. Proven ability to manage the development of optical instrumentation, from conceptual design (simulation) and CAD prototyping to research system deployment and experimental validation. Skilled in multidisciplinary technical coordination and adhering to high precision quality standards.

Research/Professional Experience

Jan 2022 to Present **Med. Univ. Innsbruck** *Innsbruck, Austria*
Systems Engineer | PhD Researcher

Closed-Loop Metrology & Wavefront Control: Designed and built from automated wavefront sensing platform integrating MEMS and phase modulators. Implemented iterative phase retrieval algorithms achieving diffraction-limited imaging performance.

R&D Prototyping & Instrumentation: Led lifecycle development of a optical imaging systems from CAD design (FreeCAD, SolidWorks) and rapid 3D-printed prototyping to hardware integration and software deployment. Developed SpectraMap, a Python package for automated ML-based classification.

DOE & Equipment Qualification: Executed structured Design of Experiments using python simulations and optical tolerance analysis to qualify adaptive optics potential performance. Performed calibration, alignment, and maintenance of laser systems, piezo stages, and scientific cameras with precision research environment.

Mar 2018 to Feb 2019 **INTECOL S.A.S.** *Medellín, Colombia*
Junior Machine Vision Engineer

High-Speed Quality Inspection: Engineered automated vision inspection systems for high-speed production lines using Halcon and Cognex. Designed custom illumination setups enabling high inspection coverage for label alignment, liquid level, and cap sealing defects.

Process Integration & Cross-Functional Support: Automated robotic surface flaming guidance for Renault bumper lines (Cognex VisionPro). Collaborated with maintenance and production teams to perform root cause analysis and implement continuous process improvements in live manufacturing environments.

Technical Competencies

Software & AI Stack: Python (JAX, scikit-learn, NumPy), LabVIEW (Real Time and FPGA), C++ (OpenCV), Halcon, Cognex, Git.

Design & Simulation: Optomechanical design in SolidWorks and FreeCAD. Optical system modelling in ZEMAX OpticStudio, including lens catalogue import and aberration analysis for system tolerancing. Rapid prototyping for experimental setups using 3D printing.

Hardware & Instrumentation: Integration of scientific cameras (basler, andor, hamamatsu), galvo scanners, piezo stages (PI), and deformable mirrors (Texas

Instruments, ALPAO, Boston Micromachines). Hands on experience with laser systems (fs, CW), fiber optics, arduino based automation systems, and DAQ cards (National Instruments).

Education

Jan 2022 to Jun 2026	Medical University of Innsbruck <i>Doctor of Philosophy in Optical Metrology</i>	<i>Innsbruck, Austria</i>
2019 to 2021	Friedrich Schiller Univ. Jena <i>Master of Science in Photonics</i>	<i>Jena, Germany</i>
2012 to 2018	University of Cauca <i>Bachelor of Science in Physics Engineering</i>	<i>Popayán, Colombia</i>

Publications/Technical Evidence Awards

Awards: Selected for ZEISS SMT, ASML, and SPIE summer schools. COLFUTURO 2019 (best Colombian professionals).

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills

Adaptability & Independent Working Style: Ability to adapt to dynamic project requirements and take on new tasks; strong focus on bringing conceptual ideas to practical applications with responsibility.

Teamwork & Process Analysis: Team oriented collaboration across optics, hardware, and software disciplines to optimize measurement workflows; structured root cause analysis of system defects and implementation of continuous improvements.

Personal Interests

- Mountain Sports (Biking, Hiking, Skiing)
- Bachata Dancing
- Harmonica
- Learning German

Languages

English (Fluent) · German (Intermediate B2) · Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte
Medical University of Innsbruck
monika.ritsch-marte@i-med.ac.at

PD Dr. Christoph Krafft
Leibniz IPHT
christoph.krafft@leibniz-ipht.de

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellin Calle 42 #63c-82
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OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garcés
TRADUCTOR INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

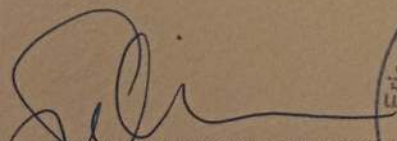
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

ASML

PhD Master Class

You will be joining us live at ASML HQ in Veldhoven on Friday and Saturday. Make sure you bring your **valid ID** to enter the ASML buildings. Without it, you will **not** be allowed inside. On Thursday we will meet at the Philips Stadium in the center of Eindhoven for an informal start to the event. The dress code for the entire masterclass is informal.

Program

Below is the program for the three days again. On the next pages, you will find more details on the different parts of the program.

Dear participant of the ASML PhD Master Class | Spring edition 2026,

We're looking forward to meeting you on March 19, 20 & 21.

Here's an information package containing details about the masterclass program. Please read this carefully so you know what to expect and how to prepare for it.

To get the most out of this masterclass for you, we want your input on what you would like to learn. At the end of this briefing, there is a link where we ask you to share your preferences with us.

Do not hesitate to contact us if you still have questions after this briefing.

Laura de Swart
Program Manager, ASML
+31 6 1562 7555

PhD Master Class

Thursday, March 19

19:30 – 22:30 Informal activity @ Philips Stadium

Friday, March 20

08:30 – 09:00	Welcome
09:00 – 10:30	Introduction into ASML Technology
10:30 – 10:50	Break
10:50 – 12:00	Pitches from former participants
12:00 – 13:00	Lunch break
13:00 – 18:00	Working on cases
18:00 – 21:45	Dinner & presentations & feedback on case results
21:45 – 22:00	Closing day one

Saturday, March 21

08:30 – 09:00	Welcome back
09:00 – 12:50	Carousel • Matching conversations • HR corner • Q&A • ASML tours
12:50 – 13:50	Lunch break
13:50 – 14:00	Wrap up and closing

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stämmler

Dr. Thomas Stämmler
CTO & Head of Product Strategy



Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ö s d

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Passaporto № / Passport No

MUNOZ BOLANOS

COLOMBIANA

Sexo / Sex
Lugar de nacimiento / Place of birth

Fecha de expedición / Date of issue

10 DIC/DEC 2028

G. ANTIOQUIA

Firma del titular / Titular y representante

Quar Family

P<COLMUNOZ<BOLANOS<<JUAN<DAVID<<<<<<<<<<<<<<<<<
AV410884<OCOL9406041M2812106CC1061772278<<56

Stadtmagistrat Innsbruck
MA II - Bezirks- und Gemeindeverwaltung
Standesamt und Personenstandsangelegenheiten
Aufenthaltsangelegenheiten
post.aufenthaltsangelegenheiten@innsbruck.gv.at

Zahl: Maglbk/58566/AUFG-NAG/3/1 – 00053337-04

Antragsprotokoll

Herr: **Muñoz Bolaños Juan David**
wohnhaft in: **6020 Pradl, Reichenauer Straße 42/Top 5**
geboren am: **04.06.1994**
Reisedokument-Nr.: **AV410884**
Nationalität: **Kolumbien**
hat am: **13.03.2026**

beim Stadtmagistrat Innsbruck, Referat Aufenthaltsangelegenheiten, einen Zweckänderungsantrag auf Rot-Weiß-Rot - Karte plus eingebracht.

Ein rechtzeitig eingebrachter Verlängerungsauftrag räumt während der gesamten Dauer des Verlängerungsverfahrens dieselbe Rechtsposition ein, die nach Erhalt des letzten Aufenthaltstitels innegehabt wurde. Dies bedeutet auch, dass bei befristet erteilten Aufenthaltstiteln mit unbeschränkten Zugang zum Arbeitsmarkt („Rot-Weiß-Rot Karte plus“ und „Familienangehöriger“) weiterhin während der gesamten Dauer des Verlängerungsverfahrens ein Zugang zum Arbeitsmarkt besteht. Bei darüber hinaus gehenden Fragen einer allfälligen weiterhin bestehenden Beschäftigung wenden Sie sich bitte an das Arbeitsmarktservice (AMS).

Nachzureichende Unterlagen: ***

Die Erledigung des vollständig eingebrachten Antrages wird voraussichtlich innerhalb von Wochen erfolgen.

Persönliche Abholung (erforderlich !) jedenfalls erst ab: 17.05.2026, Zimmer: .Nummer ziehen

Zur Abholung (Abholzeiten: Montag bis Freitag, 08.00 bis 11.30 Uhr) sind mitzubringen:

- **der Reisepass**
- **Aufenthaltstitel in Kartenform**

Für den Bürgermeister:

(Huter)

ausstellende Behörde:

Stadtmagistrat Innsbruck
Abteilung II
Aufenthaltsangelegenheiten

übernommen am: 13.03.2026

Unterschrift:

Bei überprüfenden Amtshandlungen drei Monate nach Ausstellung dieser Bestätigung wird um Rückfrage bei der zuständigen Aufenthaltsbehörde ersucht. (Tel.: 0512/5360-1030/1032/1034/1036)